```
1.
2.  #include <reg51.h>
3.
4.  // DAC connected to Port 1
5.  #define DAC P1
6.
7.  // Waveform selection buttons (active LOW)
8.  sbit SQUARE = P3^0;
9.  sbit TRIANGLE = P3^1;
10. sbit SAWTOOTH = P3^2;
11.
12. // Delay function
13. void delay(void)
14. {
15.     unsigned int i;
16.     for(i = 0; i < 100; i++);
17. }
18.
19. // Generate square wave
20. void square_wave(void)
21. {
22.     DAC = 0xFF;       // HIGH level
23.     delay();
24.
25.     DAC = 0x00;       // LOW level
26.     delay();
27. }
28.
29. // Generate triangular wave
30. void triangular_wave(void)
31. {
32.     unsigned int i;
33.
34.     // Increasing voltage
35.     for(i = 0; i <= 255; i++)
36.     {
37.         DAC = (unsigned char)i;
38.         delay();
39.     }
40.
41.     // Decreasing voltage
42.     for(i = 255; i > 0; i--)
43.     {
44.         DAC = (unsigned char)i;
```

```
45.         delay();
46.     }
47. }
48.
49. // Generate sawtooth wave
50. void sawtooth_wave(void)
51. {
52.     unsigned int i;
53.
54.     for(i = 0; i <= 255; i++)
55.     {
56.         DAC = (unsigned char)i;
57.         delay();
58.     }
59. }
60.
61. // Main program
62. void main(void)
63. {
64.     DAC = 0x00;
65.     P3 = 0xFF;        // Configure button pins as inputs
66.
67.     while(1)
68.     {
69.         if(SQUARE == 0)
70.         {
71.             square_wave();
72.         }
73.
74.         else if(TRIANGLE == 0)
75.         {
76.             triangular_wave();
77.         }
78.
79.         else if(SAWTOOTH == 0)
80.         {
81.             sawtooth_wave();
82.         }
83.
84.         else
85.         {
86.             DAC = 0x00;  // No button pressed
87.         }
88.     }
```

89. }
90.